

Real-Time Joystick Control and Experiments of Redundant Manipulators Using Cosine-Based Velocity Mapping

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Abstract—In this paper, to achieve the real-time joystick control of redundant manipulators, two schemes are proposed and investigated: one scheme is a cosine-based position-to-velocity mapping, which maps the position of the joystick in the motion range to the velocity of the robot end-effector for the real-time end-effector velocity generation; and the other is a real-time joystick-controlled motion planning (JCMP) scheme, which is based on minimum-velocity-norm (MVN) and used to resolve the velocity-assigned inverse-kinematics problem of redundant manipulators. In addition, the JCMP scheme is formulated as a standard quadratic program (QP) subject to equality and bound constraints. For the purpose of experimentation, a numerical algorithm based on piecewise-linear equations (PLE) is developed for solving the resultant QP problem. Furthermore, the proposed schemes are implemented on the planar six degrees-of-freedom (six-DOF) motor-driven push-rod robot manipulator controlled by a joystick. Computer-simulation and experiment results validate the realization and effectiveness of such schemes and the numerical algorithm.

Index Terms—Real-time motion planning, Joystick control, Piecewise-linear equations (PLE), Position-to-velocity mapping.

I. INTRODUCTION

A robot manipulator is said to be redundant when it has more degrees-of-freedom (DOF) than necessary to perform a given end-effector task [1], [2]. The fundamental issue to operate such a redundant system is the redundancy-resolution problem [1], [3], [4]. In recent decades, a favorable manner based on online optimization techniques has been applied to solving such redundancy-resolution problem [1], [5] and most of these optimization techniques can be formulated as a (time-varying) quadratic program (QP) subject to equality and inequality constraints [5], [6]. Furthermore, the constrained QP can be transformed into a system of piecewise-linear equations (PLE) [7]–[9] and then solved by a number of methods and techniques [1], [3], [5], [7]–[13].

In this paper, to achieve the real-time joystick control of redundant manipulators, a cosine-based position-to-velocity mapping scheme is firstly proposed to map the position of the joystick in the motion range to generate the desired velocity of the robot end-effector in real time. On the other hand, a real-time joystick-controlled motion planning (JCMP) scheme based on minimum-velocity-norm (MVN) is pro-

posed to resolve the velocity-assigned redundancy-resolution problem of a joystick-controlled manipulator. In addition, such schemes are implemented on the joystick-controlled redundant robot manipulator hardware system. By using the cosine-based mapping technique, the joystick can control the manipulator end-effector readily to execute many tasks.

II. PRELIMINARY AND HARDWARE SYSTEM

The joystick-computer-manipulator control system consists of a joystick, a host computer and a redundant manipulator hardware system, of which the conceptual block diagram is shown in Fig. 1. The joystick physically connects to the redundant manipulator hardware system through the host computer so that the operator can control the redundant manipulator via the joystick in real-time. Specifically, the joystick can generate and send the end-effector motion command to the host computer according to the operator's action; and, when receiving the control command, the computer and then the manipulator hardware system respond to it by performing redundancy resolution and achieving the corresponding movement and task (e.g., drawing a straight-line segment).

A. The Velocity-Assigned Inverse-Kinematics Problem

For redundant manipulators, the relationship between end-effector position vector $r \in R^m$ and joint angle vector $\theta \in R^n$ can be represented as $r = f(\theta)$ [3], [10], [14], where f is a differentiable function. Considering the resolution of the redundancy at velocity level, we get the relationship between end-effector velocity $\dot{r}$ and joint velocity $\dot{\theta}$ as

$$J(\theta)\dot{\theta} = \dot{r}, \quad (1)$$

where Jacobian matrix $J(\theta) = \partial f / \partial \theta$. It is worth pointing out here that, for the real-time joystick-controlled motion planning, the assigned end-effector path $r(t)$ is not necessary (and usually impossible) to be explicitly prescribed in mathematic equations, because the end-effector velocity $\dot{r}$ is generated according to the operator's real-time control input via the joystick directly.

B. Joystick-Controlled Manipulator Hardware System

The joystick-controlled redundant robot manipulator hardware system, which is designed for the experiment, consists

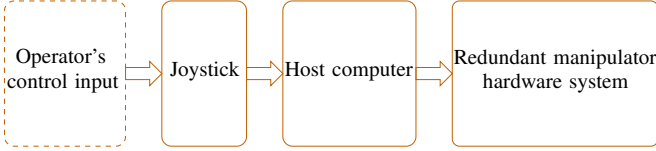
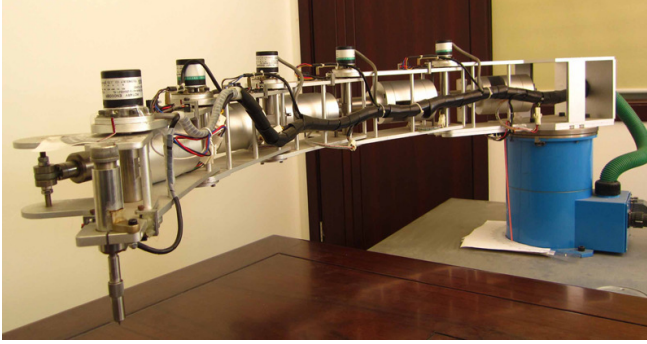


Fig. 1. Conceptual block diagram of the joystick-controlled robot system.



(a) The joystick



(b) The planar six-DOF motor-driven push-rod robot manipulator

Fig. 2. The joystick-controlled manipulator hardware system.

of three parts: a joystick, a six-DOF motor-driven push-rod manipulator, and a host computer as a linking “bridge”.

1) *Joystick*: The joystick shown in Fig. 2(a) is employed to control the redundant manipulator directly. It has an 8-way directional pad, eleven buttons (one module button and ten control buttons) and two analog sticks. One of the analog sticks is employed for the end-effector motion control, and other buttons are specially designed for auxiliary control such as joint motion initialization and joint angle adjustment. In addition, it has a USB cable, which makes the connection convenient between the joystick and the host computer.

It is worth mentioning that each button of the joystick is associated with a hexadecimal number and the host computer can read such a hexadecimal number according to each button's action. For example, when a button is pressed, we can determine which one is pressed by checking the output hexadecimal number of the joystick. In addition, the analog stick moves in a planar region, which means it needs a two-dimensional coordinate to determine its position. Thus, a planar coordinate system is established naturally (termed the original coordinate system of the analog stick). In this coordinate system, the position of the analog stick is associated with a two-dimensional coordinate (p_X, p_Y) , where

TABLE I
THE PARAMETERS USED FOR CONTROLLING THE PLANAR SIX-DOF
MOTOR-DRIVEN PUSH-ROD ROBOT MANIPULATOR

i	θ_i^- (rad)	θ_i^+ (rad)	l_i (m)	a_i (m)	b_i (m)
1	0.000	4.587	0.301	—	—
2	0.000	0.819	0.290	0.250	0.080
3	0.000	0.621	0.230	0.250	0.080
4	0.000	0.599	0.225	0.190	0.080
5	0.000	0.599	0.214	0.185	0.080
6	0.000	0.445	0.103	0.174	0.080

$p_X, p_Y \in [0, 65535]$ are obtained from the joystick. Since the analog stick's initial position is in the center of the motion space, the default initial coordinate is (32767, 32767) in such an original coordinate system. However, all the coordinate values in such an original coordinate system are positive, which may not be applicable to determining the motion direction and the displacement directly.

To remedy the shortage of the expression in the original coordinate system, we can convert (p_X, p_Y) in the original coordinate system of the analog stick to (j_X, j_Y) in a new coordinate system by the following normalization equations:

$$j_X := \frac{p_X - p_{Xmax}/2}{p_{Xmax}/2}, \quad j_Y := \frac{p_Y - p_{Ymax}/2}{p_{Ymax}/2}, \quad (2)$$

where $j_X, j_Y \in [-1, 1]$, and $p_{Xmax} = p_{Ymax} := 65535$. The initial position of the analog stick is exactly at the origin of the new coordinate system, i.e., $j_X = j_Y = 0$. Thus, once we push the analog stick, it is easy to obtain both the movement direction and the displacement via the new coordinate.

2) *Planar Six-DOF Motor-Driven Push-Rod Manipulator*: The robot shown in Fig. 2(b) is a planar six-DOF motor-driven push-rod robot manipulator. The joint physical parameters of the manipulator are displayed in Table I and used in this paper, where l_i denotes the length of the i th link, and θ_i^+ and θ_i^- denote the upper and lower limits of the i th link, respectively. The first joint is made of steel and driven by a servomotor, since it is supposed to bear the weight of the whole arm and meet the need of a wide range operational space. The second to the sixth joints are made of aluminum so as to reduce the arm weight, and driven by stepper motors.

3) *Host Computer*: The host computer is a personal computer with a Pentium(R) Dual-Core E5300 2.60GHz CPU, 4GB DDR3 memory and a Windows XP Professional operating system, which receives commands generated by the joystick and sends instructions to the manipulator motion-control module [e.g., a six-axis motion-control card of peripheral component interconnect (PCI)] to operate the redundant manipulator. Note that the host computer is physically connected to the joystick and converts the output hexadecimal number of the joystick to the manipulator control instructions (including the redundancy-resolution results). Thus, the instructions sent to the robot motion-control module originate from the joystick.

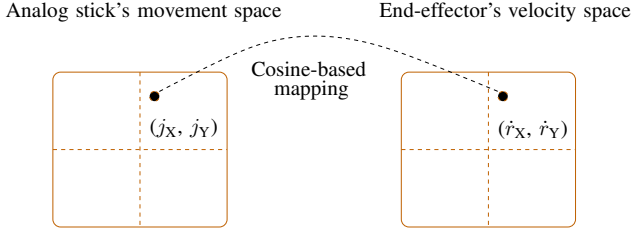


Fig. 3. Block diagram of cosine-based position-to-velocity mapping.

III. SCHEME FORMULATION AND PROBLEM SOLUTION

In this section, a cosine-based position-to-velocity mapping scheme is firstly proposed for the real-time end-effector velocity generation. With the generated end-effector target velocity, a real-time JCMP scheme subject to equality and bound constraints is proposed, which is then converted into a standard QP. Since such a QP can be equivalent to a system of PLE, a PLE-based numerical algorithm is presented to solve such a PLE problem as well as the QP problem.

A. Cosine-Based Position-to-Velocity Mapping Scheme

Since the manipulator hardware system is controlled by a joystick, a cosine-based position-to-velocity mapping scheme is proposed to map the position of the analog stick of the joystick in its motion range to the desired velocity (or to say, the target velocity, or the assigned velocity) of the robot end-effector. The conceptual block diagram of the designed mapping is shown in Fig. 3. The mapping equations are

$$\dot{r}_X := \begin{cases} j_X (1 - \cos(\pi j_X)) / |j_X|, & \text{if } j_X \neq 0 \\ 0, & \text{if } j_X = 0 \end{cases}, \quad (3)$$

$$\dot{r}_Y := \begin{cases} j_Y (1 - \cos(\pi j_Y)) / |j_Y|, & \text{if } j_Y \neq 0 \\ 0, & \text{if } j_Y = 0 \end{cases}, \quad (4)$$

where $\dot{r}_X$ and $\dot{r}_Y$ denote respectively the desired end-effector velocity components in X-axis and Y-axis, and $j_X, j_Y \in [-1, 1]$ denote respectively the relative position of the analog stick in its motion plane which are defined by equation (2). It is easy to prove that the $\dot{r}_X(j_X)$ and $\dot{r}_Y(j_Y)$ are differentiable within the range $[-1, 1]$, which demonstrates that the cosine-based position-to-velocity mapping scheme is feasible.

B. Real-Time JCMP Scheme

According to [1], [10], joint-velocity $\dot{\theta}$ in equation (1) can be obtained by the following scheme formulation with joint angle limits and joint velocity limits considered:

$$\text{minimize} \quad \|\dot{\theta}\|_2^2 / 2 \quad (5)$$

$$\text{subject to} \quad J(\theta)\dot{\theta} = \dot{r}, \quad (6)$$

$$\theta^- \leq \theta \leq \theta^+, \quad (7)$$

$$\dot{\theta}^- \leq \dot{\theta} \leq \dot{\theta}^+, \quad (8)$$

where $\|\cdot\|_2$ denotes the two norm of a vector, and superscripts $+$ and $-$ denote the upper and lower limits of a joint variable vector (e.g., θ or $\dot{\theta}$) respectively.

C. QP Formulation

Since the redundancy-resolution problem is solved at the velocity-level, the joint angle range in equation (7) could be reformulated as the following form in terms of $\dot{\theta}$:

$$\mu(\theta^- - \theta) \leq \dot{\theta} \leq \mu(\theta^+ - \theta), \quad (9)$$

where $\mu \in R$ is a positive parameter used to scale the feasible region of $\dot{\theta}$ [1], [11], [12]. Thus, equations (8) and (9) can be combined into one expression as $\xi^- \leq \dot{\theta} \leq \xi^+$, of which the i th elements (with $i = 1, 2, \dots, n$) are defined as

$$\xi_i^- := \max\{\dot{\theta}_i^-, \mu(\theta_i^- - \theta_i)\}, \quad (10)$$

$$\xi_i^+ := \min\{\dot{\theta}_i^+, \mu(\theta_i^+ - \theta_i)\}. \quad (11)$$

Based on the previous works [1], [10]–[12], the real-time JCMP scheme proposed in (5)–(8) can be reformulated as a QP subject to equality and bound constraints:

$$\text{minimize} \quad x^T W x / 2 + c^T x \quad (12)$$

$$\text{subject to} \quad Jx = b, \quad (13)$$

$$\xi^- \leq \dot{\theta} \leq \xi^+, \quad (14)$$

$$\text{with } W := I, c = 0, b = \dot{r},$$

where $x \in R^n$ denotes the $\dot{\theta}$ to be solved per sampling gap, and the i th elements of $\xi^\mp$ are given in (10) and (11).

D. Quadratic-Program Solution

Guided by [1], [10], [12], [13], solving QP problem (12)–(14) is equivalent to finding a solution to the following system of piecewise-linear equations (PLE):

$$P_\Omega(u - (Mu + q)) - u = 0, \quad (15)$$

where $\Omega := \{u \in R^{n+m} | u^- \leq u \leq u^+\} \subset R^{n+m}$,

$$u := \begin{bmatrix} x \\ y \end{bmatrix} \in R^{n+m}, u^- := \begin{bmatrix} \xi^- \\ -\varpi 1_v \end{bmatrix} \in R^{n+m},$$

$$u^+ := \begin{bmatrix} \xi^+ \\ \varpi 1_v \end{bmatrix} \in R^{n+m}, M := \begin{bmatrix} W & -J^T \\ J & 0 \end{bmatrix} \in R^{(n+m) \times (n+m)},$$

$$q := \begin{bmatrix} c \\ -b \end{bmatrix} \in R^{n+m}, 1_v := [1, \dots, 1]^T.$$

In addition, $P_\Omega(\cdot) : R^{n+m} \rightarrow \Omega$ is a projection operator, with the i th element of $P_\Omega(u)$ defined as

$$\begin{cases} u_i^-, & \text{if } u_i < u_i^- \\ u_i, & \text{if } u_i^- \leq u_i \leq u_i^+, \forall i \in \{1, 2, \dots, n+m\}. \\ u_i^+, & \text{if } u_i > u_i^+ \end{cases}$$

$y \in R^m$ is the dual decision vector defined for equality constraint (13), and $\varpi \gg 0$ is defined sufficiently large (e.g., $\varpi = 10^6$) to replace $+\infty$ numerically.

We define $\Omega^* = \{u^* | u^* \text{ is a solution of PLE (15)}\}$. According to the previous experience shown in [1], [7]–[10],

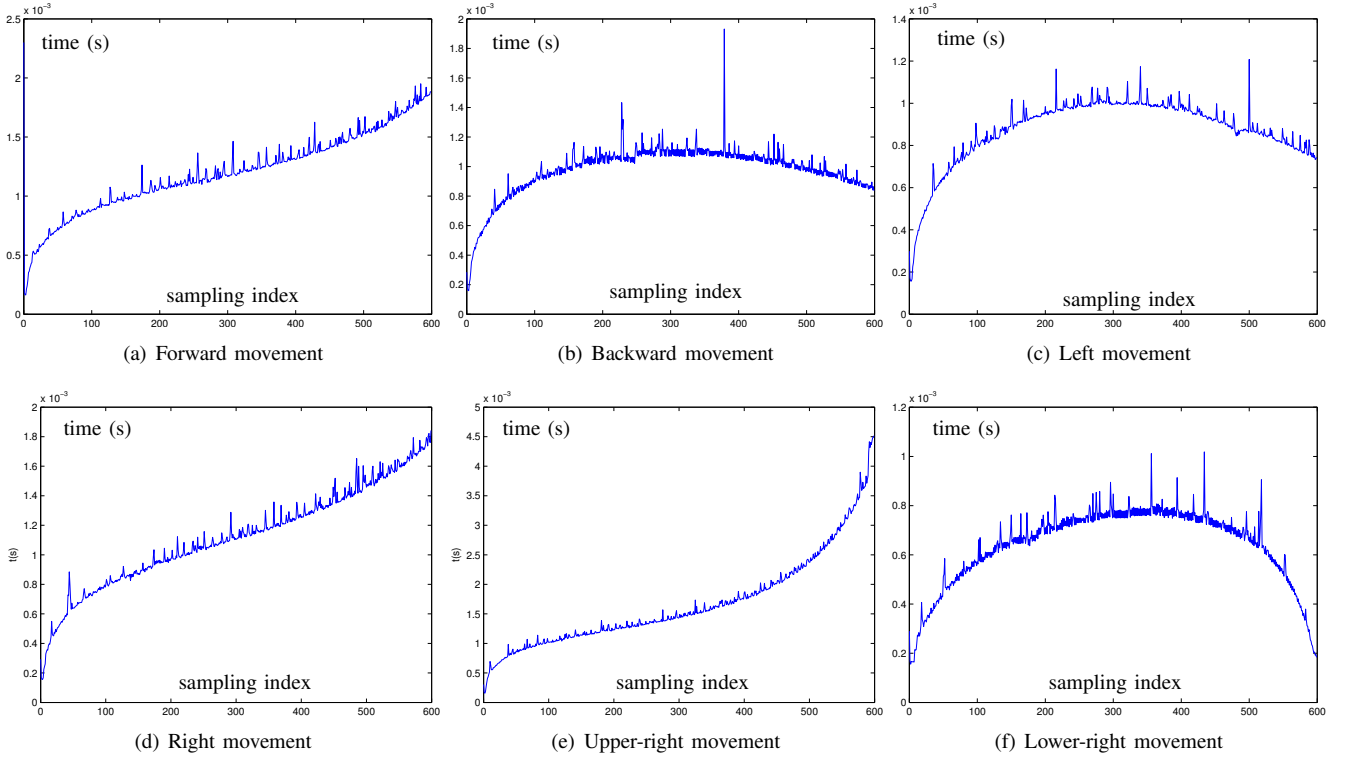


Fig. 4. The computing time used by numerical algorithm (16)-(18) for redundancy resolution in the simulations of different directions.

[13], solving (15) is equivalent to finding the zero point of the following error equation:

$$e(u) := u - P_{\Omega}(u - (Mu + q)). \quad (16)$$

By setting the initial augmented primal-dual decision variable vector $u^0 \in R^{n+m}$, we can get u^{k+1} (if $u^k \notin \Omega^*$) by the following PLE-based numerical algorithm:

$$u^{k+1} = u^k - \rho(u^k)d(u^k), \quad \forall k = 0, 1, 2, \dots, \quad (17)$$

where $\rho(u^k)$ and $d(u^k)$ are defined as

$$\rho(u^k) = \frac{\|e(u^k)\|_2^2}{\|d(u^k)\|_2^2}, \quad d(u^k) = (M^T + I)e(u^k). \quad (18)$$

Furthermore, by Theorems 1-3 of [7], sequence $\{u^k\}$ generated by PLE-based numerical algorithm (17) globally linearly converges to solution u^* , of which the first n elements constitute optimal solution x^* (i.e., $\dot{\theta}$) to QP (12)-(14).

IV. SIMULATIONS AND EXPERIMENTS

For the real-time JCMP scheme, the real-time joint velocity $\dot{\theta}$ is computed by using the numerical algorithm (16)-(18), which solves QP problem (12)-(14). In order to examine the computational efficiency of the proposed scheme, we give out the computing time in several tasks in the simulations (i.e. drawing lines toward six different directions), which is illustrated in Fig. 4. From the figure, we see that all the

computing time is less than 0.01s. Thus, we can set the sampling gap (or to say, sampling period) as $\Delta t = 0.01s$ (i.e., 10ms) in the following hardware experiments.

During the experiment, the proposed real-time JCMP scheme and the cosine-based position-to-velocity mapping scheme are implemented on the joystick-controlled redundant robot manipulator hardware system by performing two tasks, i.e., drawing straight lines toward four different directions and writing “MVN” letters in a 2-dimensional plane.

A. Movement toward Four Directions

In this experiment, the end-effector of the planar six-DOF motor-driven push-rod redundant robot manipulator is expected to move toward four directions (i.e., move forward, backward, left and right) to validate the physical realizability of the robot manipulator controlled by the joystick. Since the position of the analog stick in its motion plane is associated with the desired velocity of the end-effector via a cosine-based position-to-velocity mapping, the operator can control the velocity of the end-effector directly by the joystick. The velocity of the end-effector in X-axis and Y-axis are respectively generated by equations (3) and (4). During the real-time motion control, the proposed JCMP scheme is employed, which is solved by the PLE-based numerical algorithm (16)-(18). Without loss of generality, the tasks start with the initial joint configuration $\theta(0) = [\pi/12, \pi/12, \pi/12, \pi/12, \pi/12, \pi/12]^T \text{rad}$.



Fig. 5. Directional manipulation of the joystick.

Firstly, we push the analog stick forward as shown in Fig. 5(a), and during the task, we take snapshots of the end-effector movement by a digital camera. Then, we repeat the operation by pushing the analog stick toward other three directions, as shown in Fig. 5(b)-(d).

The actual end-effector trajectories for the four directions in the experiment are shown in Fig. 6, in which the arrows placed in the workspace indicate the expected movement directions of the end-effector. As seen from the figure, the end-effector draws four straight lines toward the four expected directions respectively, illustrating that the tasks are performed successfully. In summary, these experimental results demonstrate the realizability and effectiveness of the proposed real-time JCMP scheme and the cosine-based position-to-velocity mapping scheme for the joystick-controlled redundant robot manipulator, in addition to the efficacy of the presented PLE-based numerical algorithm on solving QP (12)-(14).

B. Letters “MVN” Writing

In this experiment, letters “MVN” are expected to be written by the end-effector of the joystick-controlled robot manipulator. The tasks are also executed with the initial joint state $\theta(0) = [\pi/12, \pi/12, \pi/12, \pi/12, \pi/12, \pi/12]^T \text{rad}$.

During the tasks, we take snapshots to get the actual trajectories of the end-effector's motion in the process of controlling the redundant manipulator via the joystick. Fig. 7 shows the controlling process of the analog stick to draw the expected letters. In addition, the end-effector actual trajectories of the three letters are shown in Fig. 8. From the figure, we can see that the end-effector can move according to the operator's control, and fulfill the letter-writing tasks well. The experimental results validate again the realization and effectiveness of the proposed real-time JCMP scheme and the cosine-based position-to-velocity mapping scheme for the joystick-controlled redundant robot manipulator, in

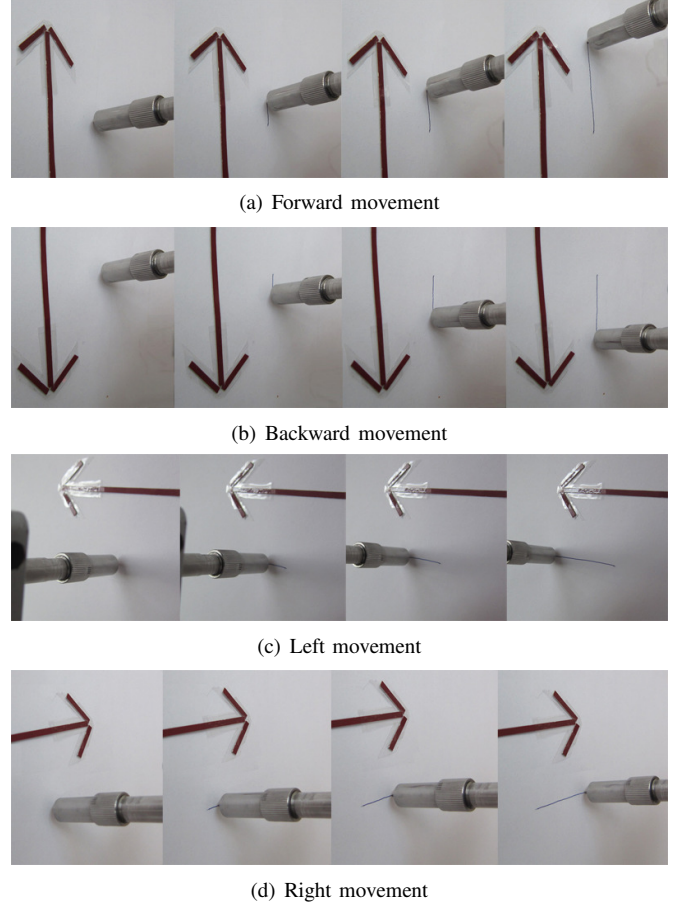


Fig. 6. The actual end-effector trajectories corresponding to Fig. 5.

addition to the efficacy of the presented PLE-based numerical algorithm on solving QP (12)-(14).

V. CONCLUSIONS

To operate the joystick-controlled redundant robot manipulator in real time, a cosine-based position-to-velocity mapping scheme has been proposed to map the position of the joystick in its motion range to the velocity of the robot end-effector (for the desired end-effector velocity generation). In addition, to achieve the real-time control of the manipulator, a real-time JCMP scheme subject to equality and bound constraints has been proposed for operating such a joystick-controlled redundant robot manipulator. The JCMP scheme can generate the desired joint velocity corresponding to the end-effector velocity given by the joystick operator. Furthermore, the proposed JCMP scheme has been solved by a PLE-based numerical algorithm. Moreover, the experimental results based on the joystick-controlled planar six-DOF motor-driven push-rod redundant robot manipulator have verified the realization and effectiveness of the proposed two schemes by executing two real-time tasks, i.e., drawing straight lines



Fig. 7. Manipulation of the joystick for writing "MVN".

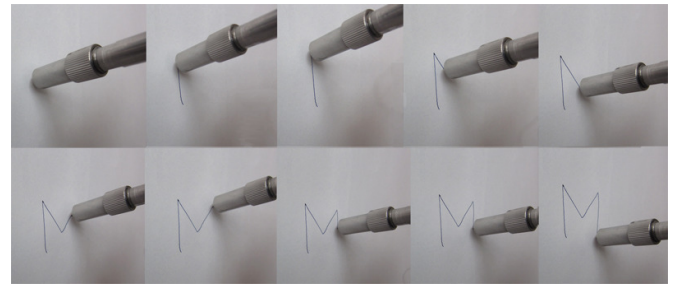
toward four different directions and writing "MVN" letters in a 2-dimensional plane.

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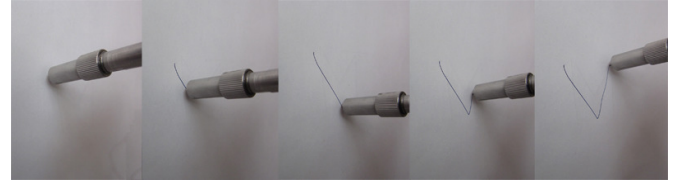
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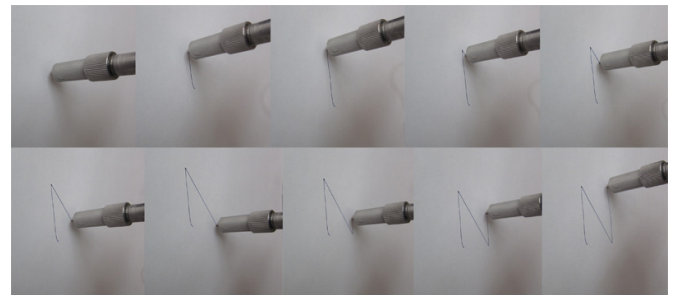
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(a) Letter "M" writing



(b) Letter "V" writing



(c) Letter "N" writing

Fig. 8. The actual end-effector trajectories corresponding to Fig. 7.